Alberto Brambila Solorzano

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Employment History

2022 – **3D CFD Simulation Engineer,** Bosch, Mexico.
• Demonstrated expertise in using industry-standard CFD software and tools, along with proficiency in surface preparation, meshing, multiphase flow, heat transfer, and turbulence modeling techniques, to support product development and optimization efforts in automotive components and systems, including E-motors, fuel cells, cooling systems, ICE.
• Developed and implemented CFD models to optimize cooling systems for internal combustion engines, resulting in a 20% increase in thermal efficiency and engine performance.
• Implemented cost-effective CFD strategies, leading to a 15% reduction in simulation software and computing expenses without compromising on analysis quality.
• Contributed to the development of innovative CFD solutions for electric vehicle (EV) battery thermal management, improving battery lifespan and overall vehicle reliability.
• Collaborated with cross-functional teams to integrate CFD findings into vehicle design, resulting in a 10% decrease in drag coefficient and enhanced vehicle stability at high speeds.

2017 – 2022 **Lecturer,** Tecnológico de Colima, Mexico.

Education

2018 – **Ph.D., Thermofluids Engineering,** UNAM, Mexico.
Modeling and numerical simulation of the dynamics of fluids in intracranial aneurysms. Using OpenFOAM, evaluate blood rheology, turbulence, and FSI models; compare the findings to rupture and nonrupture aneurysm data.

2015 - 2017 **MEng., Engineering,** Universidad de Colima, Mexico.
Computational fluid dynamics modeling and experimental investigation of a chemical vapor deposition synthesis of $\mathrm{SiC} - \mathrm{Si_3N_4}$ ***nanostructures.*** To optimize the creation of silicon nitride nanostructures, use OpenFOAM to assess the fluid dynamics and heat transfer in a reaction chamber and determine the optimal precursor location.

2009 - 2014 **BEng., Mechanical and Electrical Engineering,** Universidad de Colima, Mexico.

Skills

Technical	Computational Fluid Dynamics (CFD), Multiphase Flow, Heat Transfer, Turbulence Modeling, Aerodynamics, Thermal Management, Conjugate Heat Transfer (CHT), Spray Combustion Modeling, Post-Processing, Data Analysis, Meshing, Optimization, HPC
CAE	ANSYS Fluent, CFX, AVL FIRE M, OpenFOAM, STAR-CCM+
Coding	Python (for CFD post-processing, automation scripts), Bash (Linux), Matlab
Languages	Proficient English, native spanish
Soft	Problem-solving, communication, analytical thinking, time management, teamwork

Research Publications

1. **A. Brambila-Solórzano** et al., "Influence of blood rheology and turbulence models in the numerical simulation of aneurysms," *Bioengineering*, vol. 10, p. 1170, 2023.
2. G. M. Sánchez, C. E. Del Pozo, J. R. Medina, J. Naude, and **A. Brambila-Solórzano**, "Numerical simulation of the aqueous humor flow in the eye drainage system; a healthy and pathological condition comparison.," *Medical engineering & physics*, vol. 83, pp. 82–92, 2020.